



PRODUCT NAME

Industrial-grade Test Formula

Description

This is a cost-effective, industrial-grade test cream formulation designed for versatility and efficiency. The formulation provides a medium viscosity white cream with no added fragrance, making it suitable for a wide range of applications.

Technical Specifications

pH Level:	7.0
Viscosity:	20,000 - 40,000 cP (medium viscosity)
Shelf Life:	24 months
Batch Size:	10 - 100 kg
Processing Time:	4 hours
Temperature:	70°C during emulsification
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessel, homogenizer, heating sources
Certification:	GMP compliant

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Purified Water	Aqua	70.0%	Diluent and base for cream
2	Glycerin	Glycerin	5.0%	Humectant, moisture retention
3	Cetyl Alcohol	Cetyl Alcohol	4.0%	Emulsifier and thickener
4	Stearic Acid	Stearic Acid	4.0%	Stabilizer and viscosity control
5	Ceteareth-20	Ceteareth-20	3.0%	Emulsifier
6	Mineral Oil	Paraffinum Liquidum	7.0%	Emollient
7	Phenoxyethanol	Phenoxyethanol	1.0%	Preservative
8	Sodium Hydroxide	Sodium Hydroxide	0.5%	pH adjuster
9	Disodium EDTA	Disodium EDTA	0.5%	Chelating agent
10	Tocopherol	Tocopherol	5.0%	Antioxidant
Total			100%	

Manufacturing Process

Phase 1: Water Phase

1. Heat purified water to 70°C.
2. Add glycerin and disodium EDTA to the water and mix until fully dissolved.

Phase 2: Oil Phase

1. In a separate vessel, heat mineral oil, cetyl alcohol, stearic acid, and ceteareth-20 to 70°C.
2. Stir until all components are melted and well combined.

Phase 3: Emulsification

1. Slowly add the oil phase to the water phase while stirring continuously to form an emulsion.
2. Continue stirring until the mixture cools to 45°C.

Phase 4: Cooling Phase

1. Once the emulsion is below 45°C, add phenoxyethanol, sodium hydroxide, and tocopherol.
2. Mix thoroughly until the cream is smooth and homogeneous.
3. Check the pH and adjust with additional sodium hydroxide if necessary to reach pH 7.

Required Equipment

Mixing vessel, homogenizer, heating sources

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: GMP compliant

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10 - 100 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply the cream as needed. Avoid contact with eyes. Suitable for external use only.